

YEAR 10 / INDUSTRIAL TECHNOLOGY METAL

Folding Camping Shovel

STUDENT WORKBOOK



Name _____ Class _____

50 pages · A4 · 25 sheets duplex · Edition 1.2 · Course companion

Illustrative cover. Use teacher-issued project plans for construction.

NAVIGATION

Your learning and evidence route

Five modules, six original drawings and one connected project folio.

Learning route	Pages
Original drawing package	3–8
1. Drawings, planning and impacts	9–14
2. Shovel-head production	15–20
3. Lathe, fits and threads	21–26
4. Pivot, milling and RPM	27–32
5. Assembly and evaluation	33–38
Twelve-part project folio	39–50

Read the named theory, inspect the examples, answer every printed question and save your own evidence. Fifty original checks are selected for print; the other 100 remain online. Online numbering restarts 1–10 within each section; use the mapping and section links on the check pages. Written tasks stay beside their reading; the extended RPM record is on page 32. Continue project evidence in the matching folio.

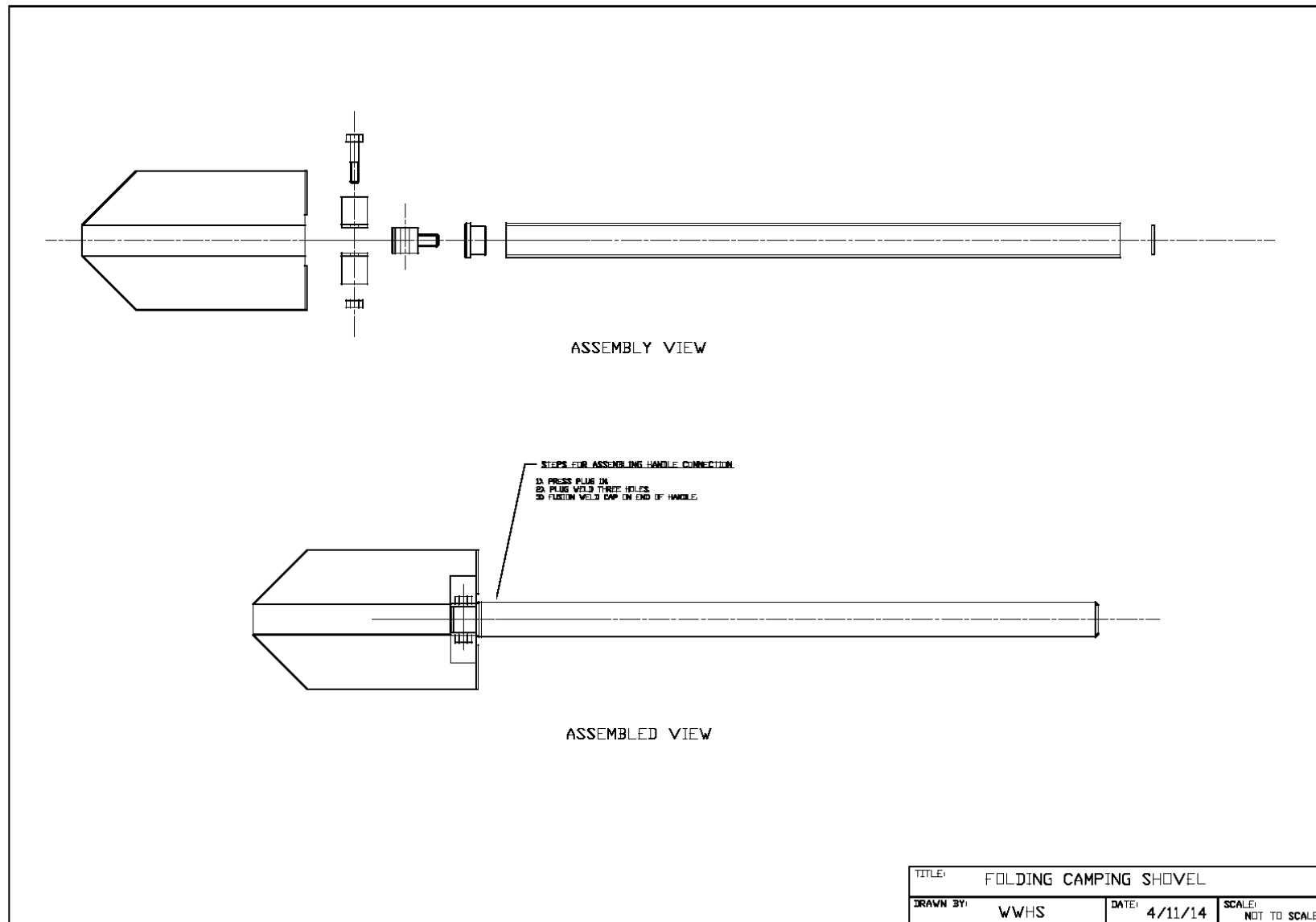
This student book is exactly 50 A4 pages, including cover and contents: 25 sheets printed double-sided on the long edge. Preserve the six landscape drawing pages and use actual-size printing. The complete source sheets retain their original notes. Never measure a resized picture to obtain a missing dimension.

Use the current issued sources The brief, drawings, SOPs, WMS and teacher demonstrations control practical details and permission. The original 2014 pivot assessment panel is historical; current assessment conditions are Teacher to confirm.

I have completed the theory, checks and written evidence, then saved or printed it as directed.

Modules 1–3	Modules 4–5
☐ Module 1	☐ Module 4
☐ Module 2	☐ Module 5
☐ Module 3	

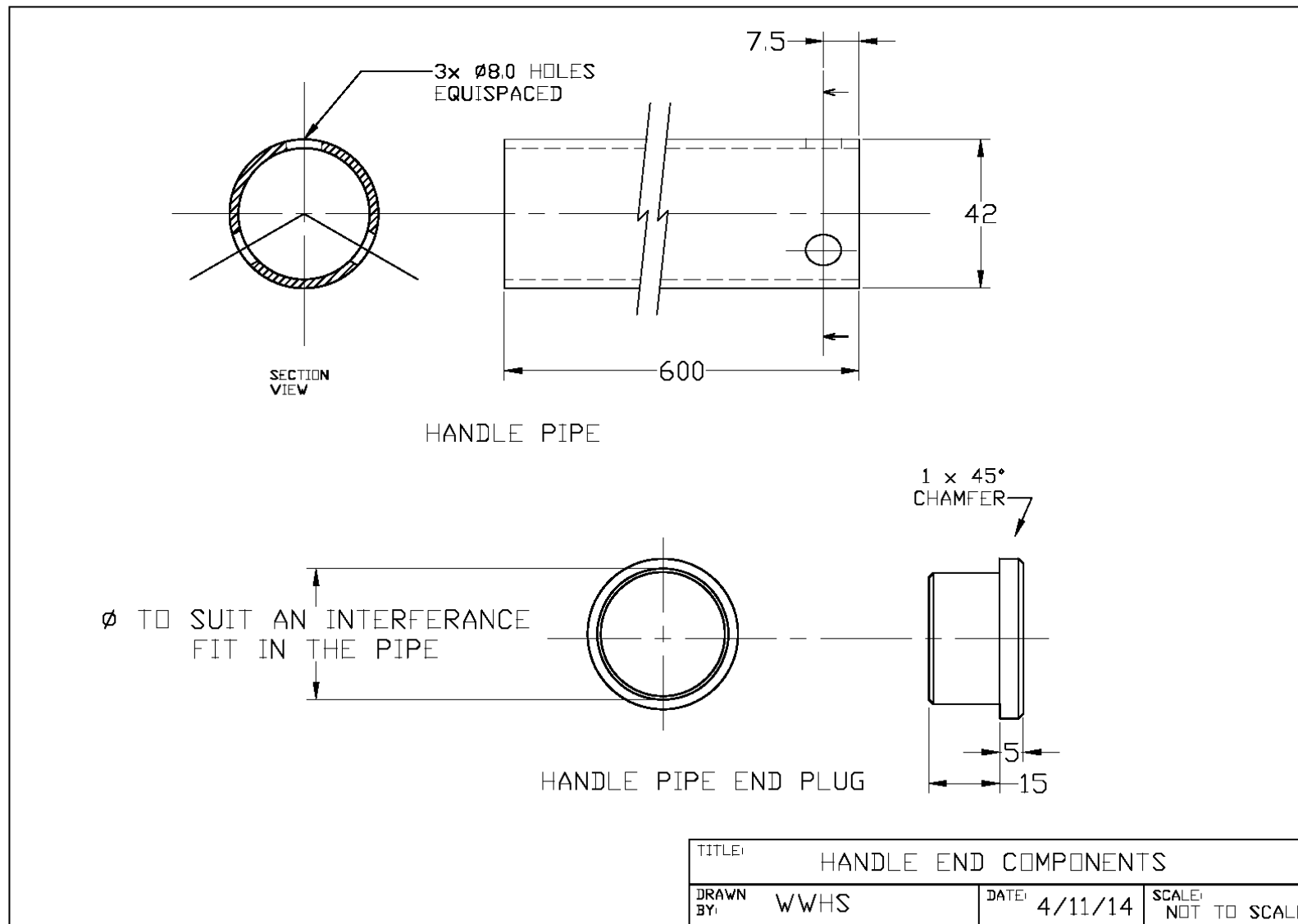
Course: stevenowell.github.io/Yr-10-Metal/folding-camping-shovel/ . Full references, image credits and answers are in the teacher guide. A checked box records readiness; it does not submit work or authorise practical activity.



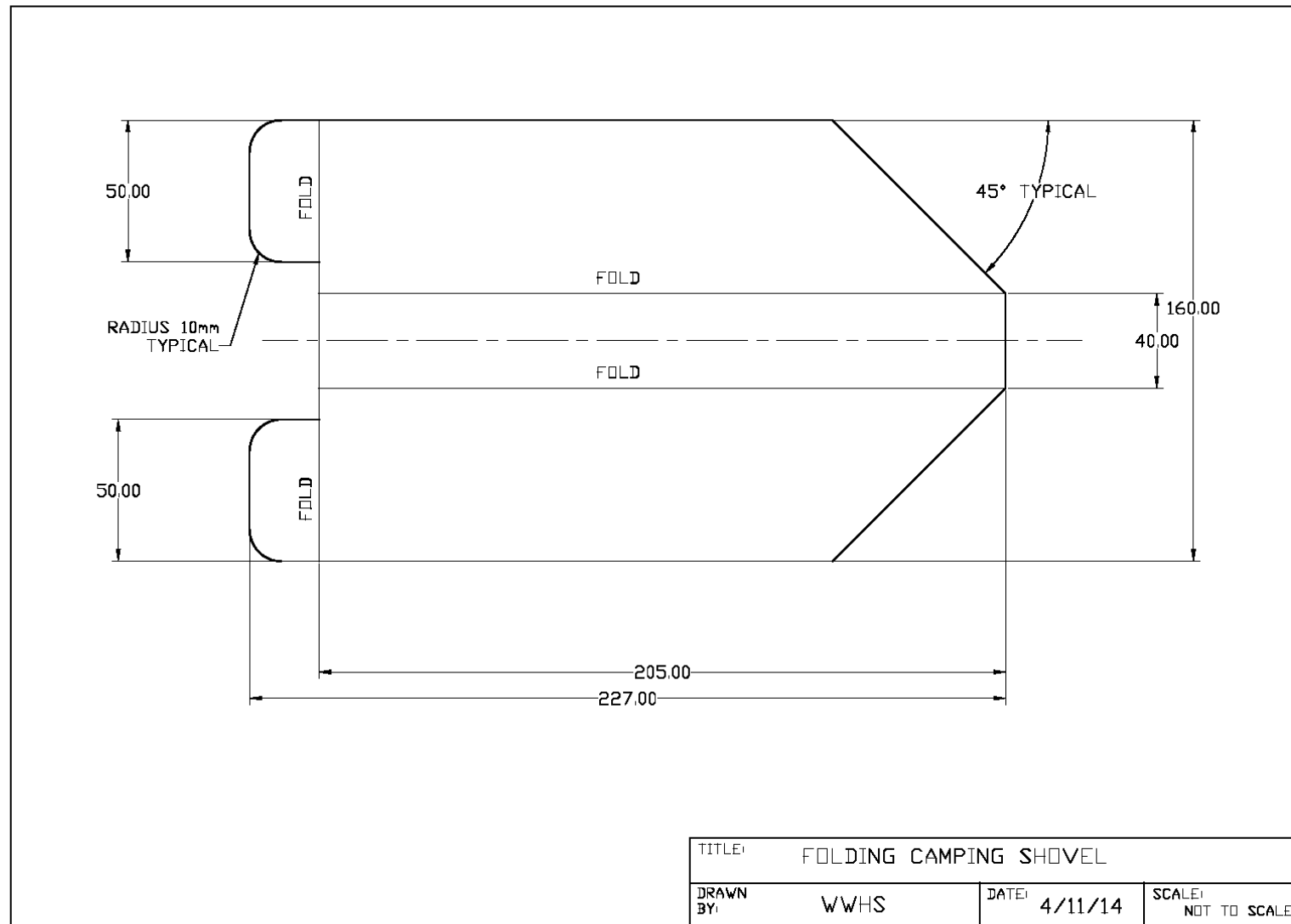
Folding Camping Shovel — assembly · Original WWHS drawing dated 4/11/14, reproduced complete and rotated for readability. Confirm the current issued version with your teacher; do not scale from this page.

Original drawing note (verbatim): STEPS FOR ASSEMBLING HANDLE CONNECTION

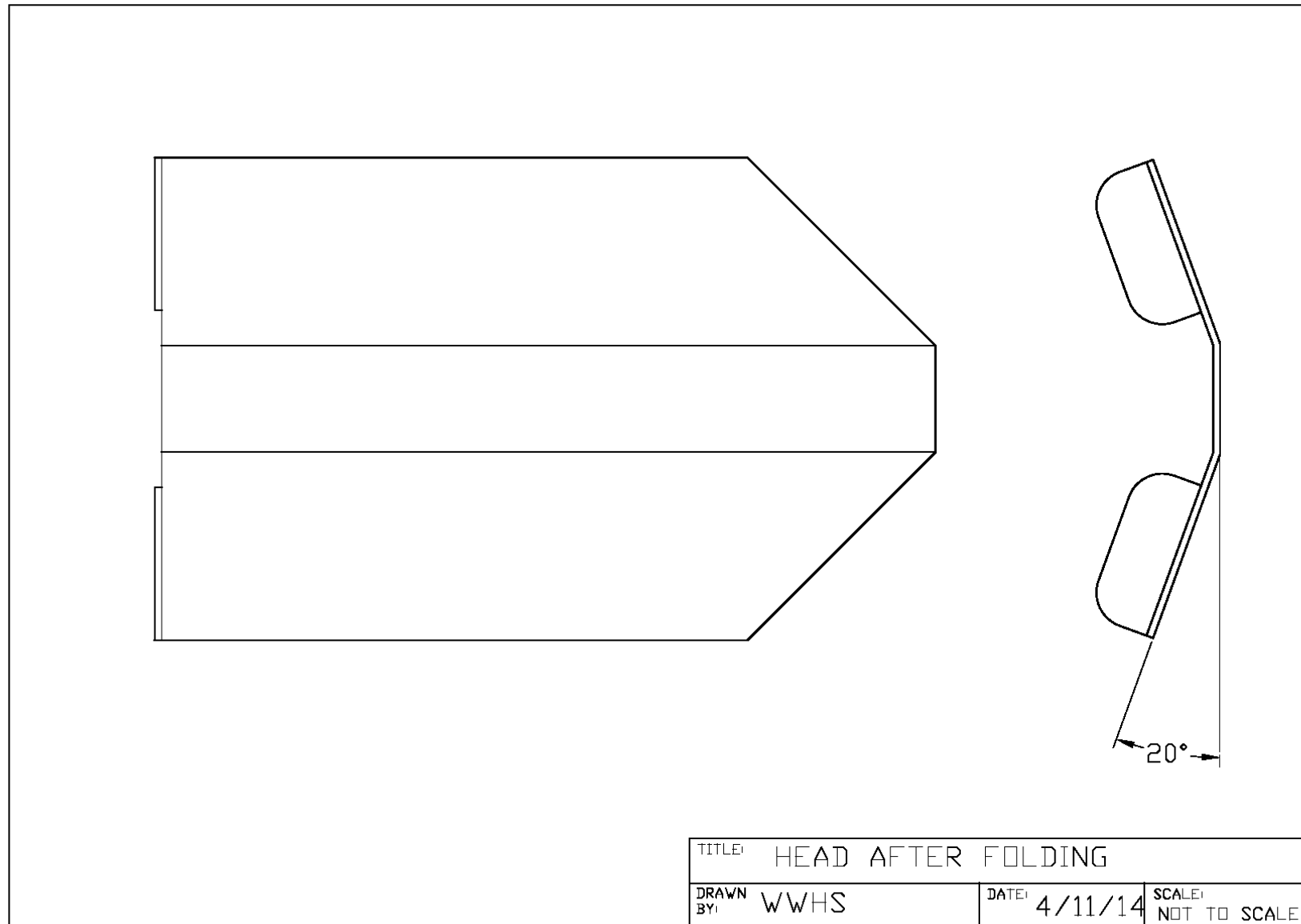
1). PRESS PLUG IN. 2). PLUG WELD THREE HOLES. 3). FUSION WELD CAP ON END OF HANDLE.



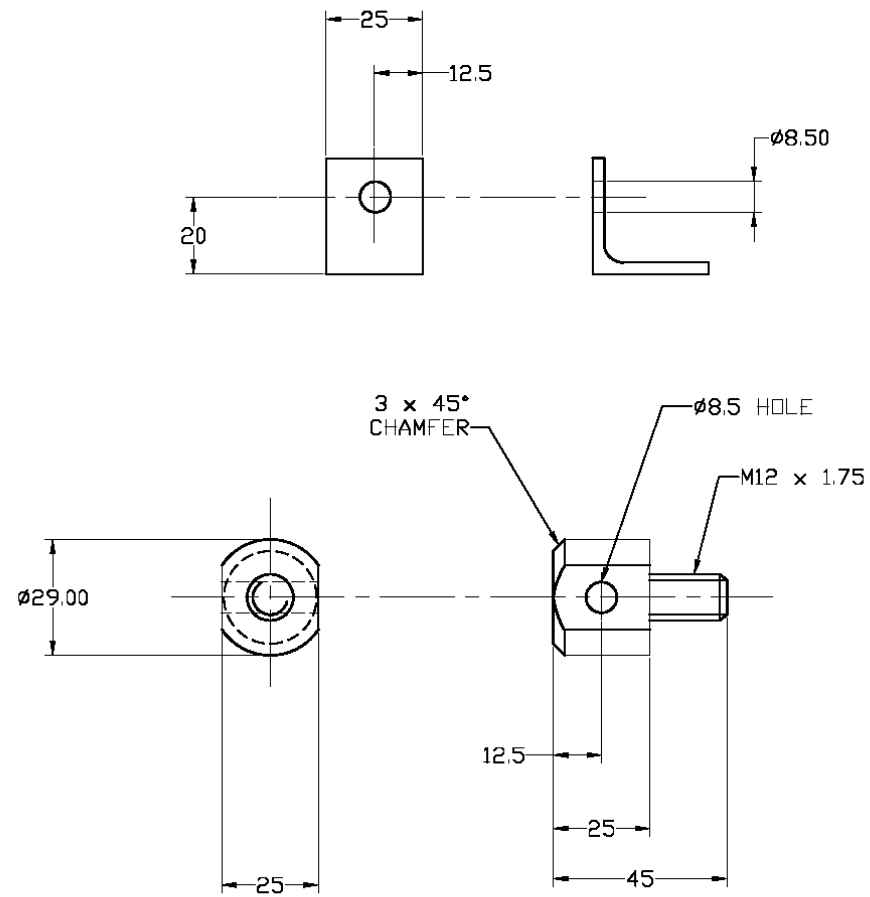
Handle End Components · Original WWHS drawing dated 4/11/14, reproduced complete and rotated for readability. Confirm the current issued version with your teacher; do not scale from this page.



Folding Camping Shovel — head flat pattern · Original WWHS drawing dated 4/11/14, reproduced complete and rotated for readability. Confirm the current issued version with your teacher; do not scale from this page.



Head After Folding · Original WWHS drawing dated 4/11/14, reproduced complete and rotated for readability. Confirm the current issued version with your teacher; do not scale from this page.



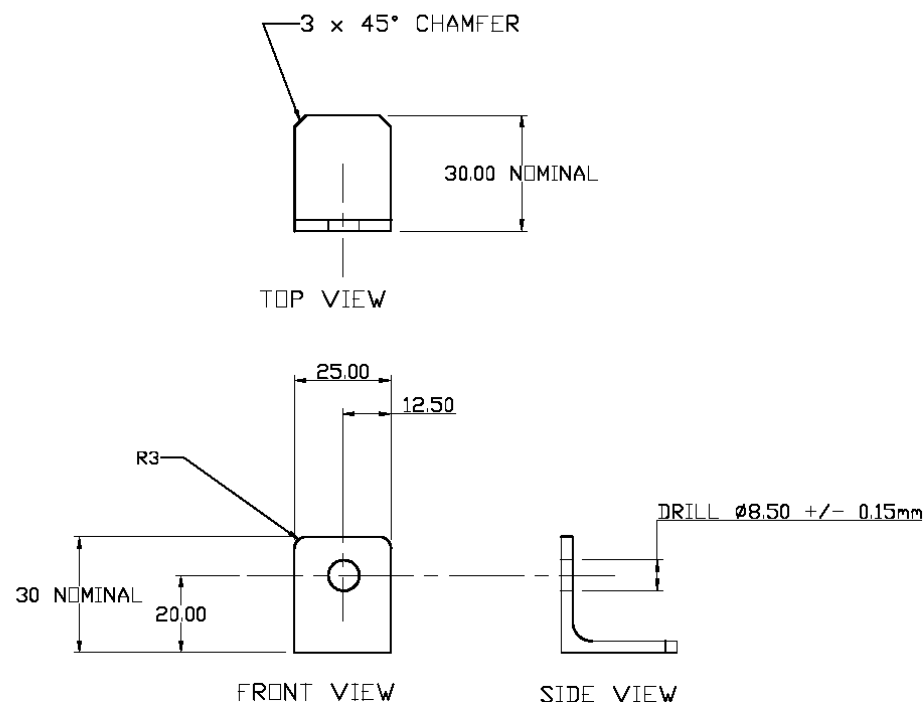
TITLE: PIVOT COMPONENTS			
DRAWN BY: WWHS	DATE: 4/11/14	SCALE: NOT TO SCALE	

Pivot Components · Original WWHS drawing dated 4/11/14, reproduced complete and rotated for readability. Confirm the current issued version with your teacher; do not scale from this page.

STUDENT NAME:**STUDENT ROLL NUMBER:****INSTRUCTIONS****TIME ALLOWED: 60 MINUTES**

1. WRITE YOUR NAME AND ROLL NUMBER ON THIS SHEET. (ABOVE)
2. FROM THE MATERIAL PROVIDED MAKE TWO PIVOT ANGLES AS PER DRAWING
3. MAKE SURE YOUR ROLL NUMBER IS STAMPED ON THE BASE OF EACH PIVOT ANGLE
4. WHEN FINISHED, TIE THE TWO PIVOT ANGLES TOGETHER WITH STRING AND SUBMIT WITH THIS SHEET FOR MARKING.

JOB FEATURE	MARKS	PART A	PART B
FILED SQUARE	4		
HOLE CENTRE 12.5 FROM SIDE	3		
HOLE CENTRE 20.00 FROM BASE	3		
HOLE DIAMETER 8.5mm +/- 0.15	2		
WIDTH 25.00	3		
3 x 45 DEGREE CHAMFER	2		
R3 RADIUS	2		
DE-BURRED	1		
CORRECT TOOLS USED	2		
WORKSHOP SAFETY	3		
TOTAL		/25	/25
FINAL MARK		/50	



PIVOT ANGLES 2 OFF
MATERIAL, 30x30x3 ANGLE

TITLE: PIVOT COMPONENTS ASSESSMENT TASK		
DRAWN BY: Wagga High	DATE: 4/11/14	SCALE: NOT TO SCALE
GENERAL TOLERANCE: +/- 0.50mm UNLESS OTHERWISE STATED		

Pivot Components: Assessment Task · Original WWHS assessment sheet dated 4/11/14. Its time and marks are historical source content. Current assessment conditions are Teacher to confirm. Do not scale from this page.

READ / OBSERVE / APPLY

Reading the folding-shovel system through its drawings

10.1 / Drawings, planning and impacts

The folding shovel is a system of related components. Its head, handle pipe, end plug and pivot must work together, so one drawing rarely answers every question. An assembly view explains where the parts belong. A detail view describes an individual part. A sectional view cuts through the imagined object to reveal information that an outside view hides.

Begin with the title and date of the current issued sheet, then identify the same component in another view. The six supplied WWHS sheets are dated 4/11/14: assembly, handle end components, head flat pattern, folded head, pivot components and the pivot-angles assessment sheet. Read them as one package. A later teacher-issued version takes precedence when your teacher confirms it.

For example, the assembly locates the handle near the pivot, while the handle detail includes a section through the pipe and an end-plug detail. Together they explain a relationship that neither view communicates completely. This does not justify inventing a plug diameter or fit allowance. A written note may require a fit to the actual pipe rather than provide a single number.

Written dimensions, symbols and notes control interpretation. A resized page is not a measuring instrument. Trace a component from assembly to detail, record what each view establishes, and flag any uncertainty before marking or machining. That trail connects the source, the physical part and your production plan, making a later decision explainable.

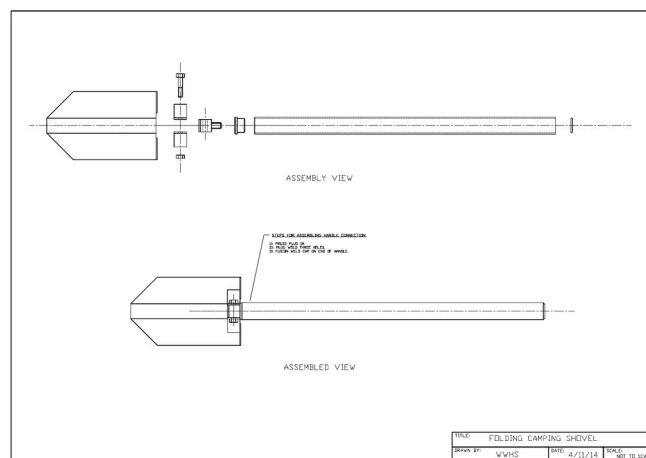


Figure 1. Source-sheet overview for locating related views. Read the complete, legible original on page 3; use its written information, not measurements from this overview.

Explain how two confirmed views/details combine to describe one component relationship.

READ / OBSERVE / APPLY

WMS planning for a multi-process product

10.2 / Connect each approved stage to the evidence it needs.

A Work Method Statement (WMS) makes the authorised production plan visible. The folding shovel involves related work on the head, handle and end plug, fits or threads, pivot, assembly and finish. A useful WMS connects these stages rather than listing tools. It records the approved task, hazards, controls, permissions and evidence needed at that point.

The connection between stages is a dependency: later work relies on an earlier result. If a component has not been checked, its uncertainty travels into the next operation. A quality hold point interrupts that chain. The work pauses until a named criterion has been compared with evidence and the teacher has confirmed the next action.

For example, a hypothetical plan may require a component check before related parts are assembled. “Check the part” is too vague. A stronger entry names the component, drawing reference, approved check, result to record and person who clears the hold point. This is an evidence model, not the shovel’s authorised making sequence.

The whole-product WMS and metal-lathe Standard Operating Procedure (SOP) have different jobs. The WMS coordinates the project; the SOP governs the approved machine process. Neither grants permission by itself. A changed drawing, setup or workpiece condition requires stopping, reporting and teacher-directed review before the plan is updated.



Figure 2. Planning evidence connects a drawing reference, the condition of a part and a recorded decision. Illustrative studio view; tools are at rest.

Justify one WMS hold point that must be cleared before a later shovel process begins.

READ / OBSERVE / APPLY

Societal and environmental criteria before manufacture

10.3 / Drawings, planning and impacts

Impact investigation begins before making because some choices become difficult to reverse after material has been processed. Start with the confirmed need and intended user. Ask how the product will be used, maintained, repaired and eventually recovered or disposed of. These questions connect its immediate function with its longer life.

Societal criteria concern people and use: whether the product meets an actual need, can be maintained, and supports responsible long-term use. Environmental criteria concern resources, processing, waste, maintenance and end of life. The categories overlap. A repairable product may remain useful longer, but the possibility of repair must be supported by its actual construction and access to suitable knowledge or parts.

A trade-off occurs when improving one criterion has a cost elsewhere. A choice intended to improve durability might require more material or processing. Using less material might reduce initial resource demand while affecting manufacture or service life. These are questions to investigate, not automatic conclusions about this shovel.

For example, replace “It is eco-friendly” with a question about a confirmed decision: what evidence would show that this choice reduces waste over the product’s life? Identify the material or process record needed, a credible external source and a limitation. Research becomes useful when it explains what could support or challenge the claim. Record uncertainty honestly; a strong report does not pretend that every decision has only benefits.



Figure 3. Resource choices have consequences before and after manufacture. Illustrative stock study; actual shovel materials remain controlled by the issued sources.

Frame one societal and one environmental research question tied to a confirmed shovel design/production decision.

MODULE 1 / SELECTED KNOWLEDGE CHECKS

Drawings, planning and impacts: checks A

Questions 2, 3, 8, 11, 13 · Circle one answer each. Read pages 9 to 11.

2. Why should students cross-reference assembly, detail and sectional views?

- A. To estimate missing dimensions
 - B. To understand complete component relationships
 - C. To avoid checking drawing revisions
-

3. Which source should be used if a written dimension and the picture appear different?

- A. The picture
 - B. The written dimension
 - C. An estimated average
-

8. If drawing information is conflicting or unclear, what is the correct action?

- A. Choose the most detailed image
 - B. Record the issue and refer it to the teacher
 - C. Scale the drawing to fill the gap
-

11. What is the main purpose of a whole-product Work Method Statement (WMS)?

- A. Coordinate the authorised construction processes
 - B. Replace teacher demonstrations
 - C. Create a new production sequence
-

13. Why are quality hold points included in a whole-product WMS?

- A. To prevent unverified work progressing
 - B. To reduce the number of processes
 - C. To replace final inspection
-

Ten original checks are printed; 20 more are online. Online numbering restarts 1–10 in each section.

Workbook Q1–10 = 10.1 Q1–10; Q11–20 = 10.2 Q1–10; Q21–30 = 10.3 Q1–10. Match section and wording.

10.1 online Q1–10 | 10.2 online Q1–10 | 10.3 online Q1–10

MODULE 1 / SELECTED KNOWLEDGE CHECKS

Drawings, planning and impacts: checks B

Questions 17, 18, 24, 26, 27 · Circle one answer each. Read pages 9 to 11.

17. If teacher-issued drawings or instructions change during the project, what should students do first?

- ☐ A. Revise the sequence independently
- ☐ B. Continue using the original WMS
- ☐ C. Stop, report the change and update the WMS only with teacher direction

18. Which statement best describes the relationship between the WMS and teacher supervision?

- ☐ A. Neither replaces teacher demonstration, supervision or authorisation
- ☐ B. The WMS replaces teacher supervision once completed
- ☐ C. Teacher supervision is only needed for lathe work

24. Which statement best reflects environmental criteria?

- ☐ A. Only the product's appearance matters
- ☐ B. Consider resources, waste, maintenance and end-of-life outcomes
- ☐ C. Estimate impacts without evidence

26. Which example best shows an evidence-based trade-off?

- ☐ A. Improving durability may require more resources or processing
- ☐ B. Choosing the fastest option is always best
- ☐ C. Reducing waste always improves service life

27. If evidence about an impact is incomplete, what should the report do?

- ☐ A. Ignore the issue
- ☐ B. Explain what evidence is still needed before reaching conclusions
- ☐ C. Replace missing evidence with assumptions

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10.1 online Q1–10 | 10.2 online Q1–10 | 10.3 online Q1–10

WORKED EXAMPLE

Turn an impact claim into an argument

Module 1 / Worked comparison and evidence investigation

An impact claim becomes useful when it connects a decision to a consequence and evidence. A statement such as “This product is good for the environment because it is metal” skips the reasoning. Material category alone cannot establish resource demand, service life, repairability or recovery at the end of use.

Weak claim	More careful reasoning
“Metal can be recycled, so this shovel has no environmental impact.”	Recycling may recover useful material, but collection, processing and manufacture still require resources. The actual material, production waste, expected service life and available recovery route need evidence.
“A durable product is always the best choice.”	A longer useful life may reduce replacement. Compare that possible benefit with extra material, processing, maintenance and the user’s actual need.

The reasoning chain Decision → likely effect → evidence needed → limitation. A supported trade-off is stronger than a broad label such as “eco-friendly”. A proposed benefit remains a possibility until evidence supports it.

Choose either weak claim above. Write a more precise research question, name two pieces of evidence needed to answer it, and explain one limitation that would stop you making a final claim.

Keep useful sources for Folio 5 (research and ideas) and Folio 11 (industry and impacts). Record who produced each source and what it can actually establish.

READ / OBSERVE / APPLY

Translating the shovel-head drawing into a controlled sequence

11.1 / Shovel-head production

A drawing becomes a production plan when each required feature is linked to a controlled action and check. First identify the shovel-head component, the issued drawing version and the written information. Compare the flat-pattern sheet with the folded-head sheet. They describe related states of the same part; neither supplies permission to choose an unconfirmed making method.

A datum is an agreed reference from which features are located or checked. Consistent references prevent a student from unknowingly measuring related features from different places. Orientation marks preserve which face or edge belongs in a particular relationship. Together, references and orientation make it possible to connect a mark on the work to a feature in the drawing.

Some features depend on others. A useful plan therefore asks what must be established and checked before later work relies on it. The teacher's production plan sets the actual sequence. Hold points belong before irreversible stages and before an error could be carried into another feature. A list of operations without these connections is an incomplete plan.

For example, a plan entry can name a verified fold feature, its drawing reference, the authorised operation category, the required check and the evidence to save. It should not invent a forming allowance or tool setting. Clear component labels, dated observations and authorised photographs provide traceability: another reader can follow the reasoning from source to developing part and folio record.

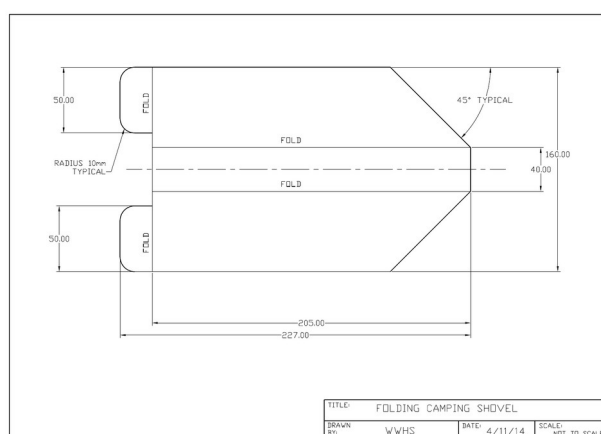


Figure 4. Source-sheet overview for locating related views. Read the complete, legible original on page 5; use its written information, not measurements from this overview.

Convert one confirmed drawing feature into a plan entry containing reference, operation category, check and evidence.

READ / OBSERVE / APPLY

Material behaviour during forming and shaping

11.2 / Shovel-head production

Forming changes shape through controlled force. Material properties influence the response, but a property name does not predict every result. Ductility concerns deformation without failure, particularly under tensile loading; malleability concerns deformation under compressive loading. Hardness concerns resistance to indentation and may affect further deformation or cutting. Thickness also influences resistance to a change of shape.

Manufacture can give material a directional character. Where relevant to the confirmed stock, orientation may affect forming consistency. Repeated deformation can cause work hardening: the material becomes more resistant to further deformation. Springback is a partial return towards the earlier shape when the applied force is removed. These are concepts to recognise, not instructions to compensate by an invented amount.

Watch the actual work during a teacher-approved demonstration or process. Record changes in shape, resistance and surface condition, then compare them with the drawing and agreed criterion. An unexpected shape may have more than one explanation. The observation alone cannot prove which property or process caused it.

For example, “the shape changed after the force was removed” is an observation consistent with springback. Before proposing an adjustment, identify the confirmed material, process and check that would distinguish this from a reference or measurement error. Report the evidence for teacher review. The optional malleability and ductility video reinforces these terms; the current SOP and demonstration still control practical action.



Figure 5. Observe form and surface condition before explaining a result. Illustrative material study; appearance alone does not establish a cause or an approved correction.

Explain one observed material response and the evidence needed before deciding on a teacher-approved adjustment.

READ / OBSERVE / APPLY

Checking head geometry, edges and assembly readiness

11.3 / Shovel-head production

The shovel head is ready for the next stage only when relevant checks support that decision. Inspection compares the developing component with the current drawing and teacher-confirmed criteria. Appearance can alert you to a possible issue, but a tidy-looking part may still have the wrong shape, feature position or relationship to another component.

Check geometry, symmetry where it is specified, feature location, distortion, and surface and edge condition. These features interact. Distortion may change alignment; an unsuitable edge condition may interfere with fitting. Use the authorised measurements or observations for the feature being checked, and identify the component and stage in the record.

A useful record separates criterion, evidence, result and response. The criterion states the requirement. Evidence describes what was actually measured or observed. The result compares the two. A possible cause is a hypothesis, not another observation. The response records the teacher-confirmed action and any recheck; it does not hide the original finding.

For example, “a feature is misaligned with the confirmed reference” describes a finding. Earlier marking, orientation, forming or handling might be investigated, but the record must show what supports any cause. Do not pass the head simply because the next components are ready. A hold point protects later work from a defect that could become harder to diagnose after assembly.



Figure 6. Condition, geometry and the source criterion belong in the same inspection record. Illustrative close-up; your own component supplies the actual evidence.

Complete a quality record separating criterion, inspection evidence, result, possible cause and authorised next step.

MODULE 2 / SELECTED KNOWLEDGE CHECKS

Shovel-head production: checks A

Questions 3, 5, 8, 11, 15 · Circle one answer each. Read pages 15 to 17.

3. What is the main purpose of datums during manufacture?

- A. To decorate the drawing
 - B. To estimate finished appearance
 - C. To provide consistent reference points for locating and checking features
-

5. Why should students identify feature dependencies?

- A. To avoid recording evidence
 - B. To recognise that some features depend on earlier verified work
 - C. To replace the approved production plan
-

8. What should students do if they find uncertainty or a mismatch?

- A. Ignore it if most features match
 - B. Refer it to the teacher before continuing
 - C. Correct it using assumptions
-

11. Which property describes a material's ability to change shape without failing?

- A. Hardness
 - B. Ductility
 - C. Thickness
-

15. What is work hardening?

- A. A material becoming more resistant to further shaping after repeated deformation
 - B. A material permanently returning to its original shape
 - C. A material becoming softer after repeated deformation
-

Ten original checks are printed; 20 more are online. Online numbering restarts 1–10 in each section.

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11.1 online Q1–10 | 11.2 online Q1–10 | 11.3 online Q1–10

MODULE 2 / SELECTED KNOWLEDGE CHECKS

Shovel-head production: checks B

Questions 16, 17, 21, 24, 26 · Circle one answer each. Read pages 15 to 17.

16. What is springback?

- A. A surface becoming harder after cutting
- B. A material partly returning towards its earlier form after shaping force is removed
- C. A workpiece cracking during forming

17. A material behaves differently from expected during approved work. What should students do first?

- A. Apply greater force immediately
- B. Assume the material is faulty
- C. Record the response and refer it to the teacher

21. What should the completed shovel head be compared with during inspection?

- A. Its overall appearance
- B. The teacher-issued drawing and confirmed quality criteria
- C. Other completed components

24. What should be recorded before suggesting a possible cause of a problem?

- A. The observed fault
- B. The expected correction
- C. The next assembly step

26. If inspection results are uncertain, what should happen before assembly continues?

- A. Continue because later parts are ready
- B. Ignore the uncertainty if geometry looks acceptable
- C. Document the doubt and refer it to the teacher

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11.1 online Q1–10 | 11.2 online Q1–10 | 11.3 online Q1–10

INVESTIGATE / APPLY

Audit a head-production record

Module 2 / Distinguish a finding from an unsupported conclusion.

The following three statements are hypothetical records, not observations of your own shovel. An audit asks what each record establishes and what still needs checking. Use the drawing and inspection principles before deciding which statement is strongest.

Record	Hypothetical inspection note
A	The head looks neat, so it is ready for assembly.
B	An edge was observed to differ from the confirmed reference. The component and inspection stage were recorded; the cause remains for review.
C	The head is distorted because the material is wrong. A correction was chosen without checking the source or teacher demonstration.

For A, B and C, identify the strongest evidence present and the most important missing information or unsupported claim. Explain which record provides the best starting point for a teacher review.

Rewrite either A or C as an honest provisional inspection record. Include a source criterion, the evidence still needed and the next teacher checkpoint; do not invent a completed check.

LOOK CLOSELY

Metal-lathe SOP evidence for handle and end-plug work

12.1 / Lathe, fits and threads

Explain the demonstrated controls

Metal-lathe SOP evidence should explain understanding, not merely copy a procedure or collect a signature. In the handle and end-plug work, connect the teacher-confirmed machining purpose to the correct drawing and the current Standard Operating Procedure. The record should identify the component or feature as well as the procedure used.

During the demonstration, recognise the important control points. They may concern permission, the approved setup, workpiece security, clearances, controlled use, stopping and inspection. Explain why two actual controls matter in that demonstrated context. The exact actions come from the local SOP and teacher; this workbook does not supply substitute operating steps.

Record what the evidence shows

A control and its evidence are different. Naming a requirement says what should happen. An annotated observation or authorised photograph shows what was observed. For example, a caption should identify the demonstrated control, its purpose and what the evidence actually shows, without claiming that a still image proves every action was performed correctly.

Approval applies to the demonstrated conditions. A changed drawing, workpiece, machine, setup or instruction needs reporting and review; do not assume earlier permission continues. Inspection closes the chain by comparing the machined feature with the current drawing and acceptance criteria before later fitting. Preserve both the result and any teacher review so the folio explains what happened, why it mattered and what remained unresolved.



Figure 7. A machined component is evidence of an outcome, not proof that every control was followed. Link the actual process record to the drawing and current SOP.

Explain the purpose of two teacher-confirmed SOP controls and the evidence that they were applied.

LOOK CLOSELY

Functional fits connect dimensions to movement and assembly

12.2 / Lathe, fits and threads

Choose the intended relationship

A fit is a dimensional relationship between parts that must assemble or move together. A clearance fit leaves space between related parts. A transition fit may give slight clearance or slight interference. An interference fit produces a tight dimensional relationship. These terms explain possible behaviour; they do not identify the required fit everywhere in the shovel.

The drawing and acceptance criteria establish which relationship is intended. The handle end-plug sheet specifically calls for an interference fit to the pipe. It does not supply a universal allowance that can be copied onto another component. The actual mating parts, current source and teacher-confirmed measurement method matter.

Verify the fit

A suitable fit can influence movement, alignment, unwanted looseness, wear and serviceability. This is why machining is more than producing one isolated number. The size of one component has meaning in relation to the part that receives it. Progressive checks allow a developing feature to be reviewed before additional material removal or assembly makes a problem harder to resolve.

For example, a record that lists only the outside appearance cannot establish a fit. A stronger record identifies both components, their sourced requirements, approved measurements and the evidence supporting the judgement. Unexpected movement or assembly difficulty is a reason to document and review the relationship, not to force the parts or choose a correction from appearance. Keep the proposed cause separate from the verified finding.



Figure 8. Fit concerns the relationship between mating parts. This illustrative detail does not specify a project allowance or demonstrate an accepted fit.

Explain how drawing values and measurements would support a teacher-confirmed fit judgement for one nominated pair.

LOOK CLOSELY

Threads as controlled geometry and evidence

12.3 / Match the specified geometry, then record the check.

A thread is a repeated form that lets matching components engage. An external thread is on the outside of a part; an internal thread is inside an opening. The crest is the high part of the profile and the root is the low region between crests. Pitch is the distance between corresponding points on neighbouring forms, measured along the thread axis.

Major diameter is the largest thread diameter and minor diameter the smallest. These names describe geometry; they do not prove that two parts match. Form, size and condition all matter. The original pivot sheet has a written thread callout. Read that callout in its component context rather than choosing an apparently similar fastener.



Figure 9. A thread's repeated crests and roots can be observed closely. This illustrative detail is not a gauge or a substitute for the specified check.

Inspection links the identified component and specification to a teacher-approved gauge or checking method. Record what was checked, the evidence, the criterion and the result. A thread that looks clean can still have the wrong geometry. Damage, contamination or incomplete form may affect engagement; resistance alone does not identify the cause.

Observation before diagnosis “Resistance was observed” describes evidence. “The pitch is wrong” is a proposed cause that needs checking. Record the symptom and refer it for teacher review; do not force a fit or invent a correction.

Produce an inspection explanation showing how a specified thread would be identified, checked and recorded using teacher-approved evidence.

MODULE 3 / SELECTED KNOWLEDGE CHECKS

Lathe, fits and threads: checks A

Questions 1, 3, 10, 11, 14 · Circle one answer each. Read pages 21 to 23.

1. What should metal-lathe SOP evidence primarily demonstrate?

- A. Understanding of the authorised process
 - B. A copied SOP document
 - C. A completed signature
-

3. Where must exact machining actions come from?

- A. Previous workshop experience
 - B. A generic lathe guide
 - C. The current SOP and teacher demonstration
-

10. If equipment differs from the demonstrated setup, what should students do first?

- A. Adapt the process independently
 - B. Continue if the component is unchanged
 - C. Report the changed condition before proceeding
-

11. What does a functional fit describe?

- A. The surface appearance of a component
 - B. The dimensional relationship between components that assemble or operate together
 - C. The machining sequence for producing a component
-

14. What should determine the required fit for the shovel project?

- A. The student's preferred fit category
 - B. The teacher-issued drawing and acceptance criteria
 - C. The easiest fit to machine
-

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12.1 online Q1–10 | 12.2 online Q1–10 | 12.3 online Q1–10

MODULE 3 / SELECTED KNOWLEDGE CHECKS

Lathe, fits and threads: checks B

Questions 17, 19, 21, 25, 26 · Circle one answer each. Read pages 21 to 23.

17. Which statement correctly describes a transition fit?

- A. It always provides space between components
 - B. It always requires tight engagement
 - C. It may produce either slight clearance or slight interference
-

19. What should a student do if unexpected movement or assembly difficulty is found?

- A. Remove more material immediately
 - B. Document the issue and refer it to the teacher before attempting correction
 - C. Ignore the problem if the parts still fit together
-

21. What does the pitch of a thread describe?

- A. The distance between corresponding points on neighbouring thread forms
 - B. The largest diameter of the thread
 - C. The lowest area between adjacent crests
-

25. What is the purpose of using a teacher-approved thread gauge or checking method?

- A. To compare the thread with the drawing and acceptance criteria
 - B. To make any thread fit during assembly
 - C. To estimate the thread size by feel
-

26. If a thread shows resistance during engagement, what should a student do?

- A. Force the components together
 - B. Document the issue and refer it to the teacher before assembly
 - C. Replace it with a similar-looking fastener
-

Ten original checks are printed; 20 more are online. Online numbering restarts 1–10 in each section.

Workbook Q1–10 = 12.1 Q1–10; Q11–20 = 12.2 Q1–10; Q21–30 = 12.3 Q1–10. Match section and wording.

12.1 online Q1–10 | 12.2 online Q1–10 | 12.3 online Q1–10

DRAW / EXPLAIN

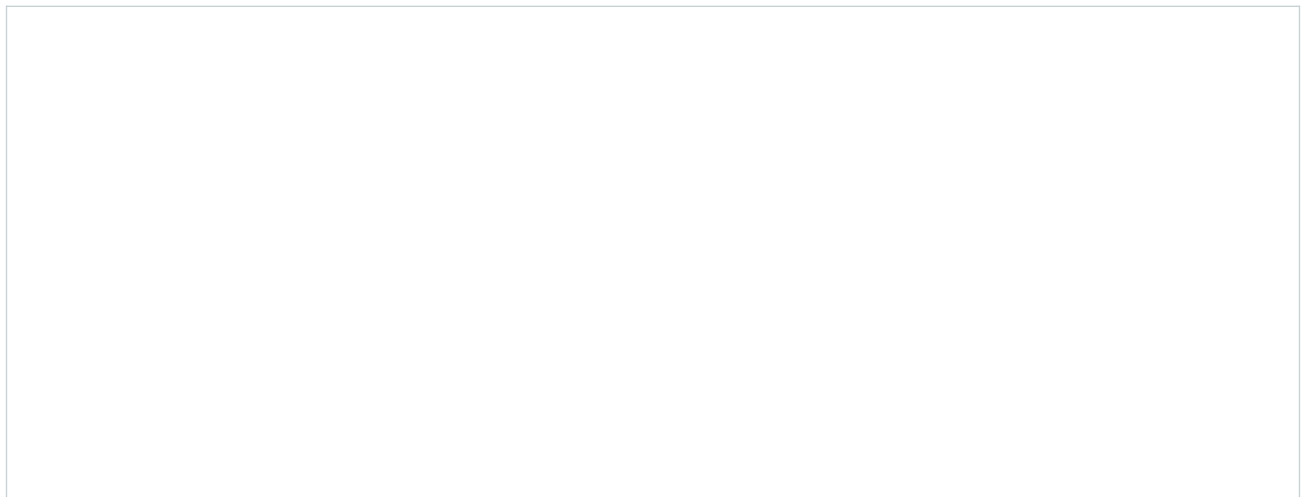
Diagnose before changing a part

Module 3 / Fits and threads require evidence, not force.

In this hypothetical inspection, two related parts do not engage as expected. The first note says: 'The parts resist movement, so the thread must be wrong.' That combines an observation with an unproven cause. More than one feature or condition may explain the same symptom.

A useful diagnostic sequence Start by identifying the parts and their current specification. Separate the observed resistance from possible causes such as mismatched specification, condition, incomplete geometry or another assembly relationship. An approved check can provide evidence; appearance alone cannot. Record uncertainty and seek teacher review before any correction.

Draw an evidence map with the observed symptom in the centre. Add three possible causes as branches and write one teacher-approved type of check or source needed to investigate each. Mark any unconfirmed method 'Teacher to confirm'.



Explain why matching one visible feature does not prove a functional fit. State what must be established before either component is altered or the assembly proceeds.

LOOK CLOSELY

Pivot parts: centres, alignment and movement path

13.1 / Pivot, milling and RPM

Locate the pivot

A pivot centre defines where related components rotate. Its position affects the path of movement and how surrounding parts interact. The drawing's written centre references therefore matter as much as the outer shape. Trace the relevant centre across the original pivot and assembly sheets instead of estimating its position from a photograph.

Where features must align, consistent datums and orientation preserve their intended relationship. Small differences between several parts can accumulate. Contact faces also matter: their condition and alignment influence how parts sit together. A component can look acceptable alone yet reveal a mismatch when compared with its related parts.

Check clearance and movement

Clearance is the available space between parts. Too much or too little may change alignment or movement, but the required amount must come from the issued drawing and teacher criteria. Do not invent a gap because a generic mechanism uses one. Similarly, an attractive image cannot establish an authorised locking action or operating position for this shovel.

Dry assembly is a reversible diagnostic stage carried out only as demonstrated. It checks identity, orientation, contact, alignment and the controlled movement path before finishing or final locking. If movement binds, record where and under what confirmed conditions it occurs. Compare that observation with measured or visual evidence before proposing a cause. Teacher review determines the next action; the purpose of the check is to reveal uncertainty while it can still be investigated.

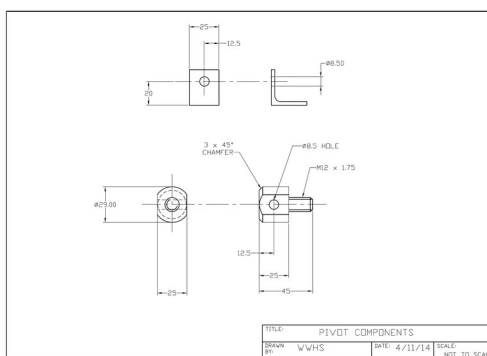


Figure 10. Source-sheet overview for locating related views. Read the complete, legible original on page 7; use its written information, not measurements from this overview.

Explain a dry-assembly check that traces one observed movement issue back to measured or visual evidence.

LOOK CLOSELY

Milling as a controlled relationship between tool, work and reference

13.2 / Pivot, milling and RPM

Connect the cutter, work and reference

Milling removes material through the relative movement of a rotating cutter with multiple cutting edges and a controlled workpiece. The outcome depends on the relationship between tool, work and reference. Simply bringing a cutter into contact with material does not define a valid setup, movement or finished feature.

A datum links the operation to the drawing. It provides a consistent basis for locating the work and checking where a feature develops. If the reference changes or the workpiece moves, a measurement may no longer refer to the intended location. Accurate results therefore depend on maintaining the demonstrated relationship, not only reading a scale carefully.

Confirm the setup and inspection

The exact workholding, clearances and movement path come from the current SOP and teacher demonstration for that job. A setup seen elsewhere is not automatically suitable. Authorisation is tied to the confirmed drawing, workpiece, equipment and process. Any change or unexpected result requires reporting and review before continuing.

Progressive inspection connects the developing feature back to its requirement before more material is removed. A traceable record identifies the drawing feature, confirmed reference, authorised setup and inspection hold point. For example, the record can explain why a check is needed before a later operation depends on the feature. It must not supply an unconfirmed feed, cutter, depth, tool path or correction. This turns machine observation into evidence of controlled decision-making.



Figure 11. The record connects the drawing reference to a confirmed process and inspection result. Illustrative inspection context; no machine setup is prescribed.

Describe how a drawing reference, confirmed setup and inspection hold point form a traceable milling record.

LOOK CLOSELY

Drilling RPM calculations: formula, units and authorised setting

13.3 / Pivot, milling and RPM

Spindle speed describes revolutions per minute (RPM). Cutting speed describes the distance travelled at the cutting edge in one minute. The two are related, but they are not interchangeable numbers. For the same cutting speed, a larger drill diameter travels a greater distance per revolution and therefore requires fewer revolutions per minute in the calculation.

Metric arithmetic relationship

$$n = (1000 \times vc) \div (\pi \times D)$$

| n: RPM | vc: m/min | D: mm. Theoretical speed and authorised setting are different records.

For metric arithmetic practice, $n = (1000 \times vc) \div (\pi \times D)$, where n is RPM, vc is cutting speed in metres per minute and D is drill diameter in millimetres. The factor 1000 converts metres to millimetres. The denominator $\pi \times D$ is a circumference. Using consistent units allows distance per minute to become revolutions per minute.

Illustrative worked substitution

Illustrative example only: with $vc = 20$ m/min and $D = 10$ mm, $n = 20,000 \div (\pi \times 10) \approx 636.6$ RPM.

These invented practice inputs are not the shovel's material data, a drill choice or an instruction to set a machine. They demonstrate substitution and units. Follow your teacher's rounding direction for actual work.

A theoretical result is not an authorised setting. For the nominated operation, confirm the formula, current cutting-speed data and drill diameter with the teacher, then compare the calculated value with available machine settings through the current SOP and demonstration. Record the approved choice and who confirmed it. Missing values, mixed units or an unexplained setting require clarification before work proceeds.

In the practice example, explain what the factor 1000 does and why the calculated value cannot by itself authorise a machine setting.

Complete the practice and the exact source calculation task in the larger record on page 32.

MODULE 4 / SELECTED KNOWLEDGE CHECKS

Pivot, milling and RPM: checks A

Questions 1, 3, 8, 11, 14 · Circle one answer each. Read pages 27 to 29.

1. What does the pivot centre primarily control in the folding mechanism?

- A. The relative movement path of related components
 - B. The finished surface appearance
 - C. The order of assembly
-

3. Why are consistent datums and orientation marks important when features must align across several parts?

- A. They reduce the need for inspection
 - B. They improve the component's appearance
 - C. They help preserve the intended relationship between matching features
-

8. Why is dry assembly completed before finishing or final locking?

- A. To permanently secure the assembly
 - B. To check identity, orientation, contact, alignment and movement while changes are still reversible
 - C. To remove the need for later inspection
-

11. According to the theory, what does milling primarily control?

- A. The relationship between the rotating cutter and the secured workpiece
 - B. The speed at which material disappears
 - C. The appearance of the finished surface
-

14. A student notices the setup differs from the teacher demonstration. What should they do?

- A. Continue if the setup looks similar
 - B. Copy another student's setup instead
 - C. Stop and follow the teacher-approved setup and demonstration
-

Ten original checks are printed; 20 more are online. Online numbering restarts 1–10 in each section.

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13.1 online Q1–10 | 13.2 online Q1–10 | 13.3 online Q1–10

MODULE 4 / SELECTED KNOWLEDGE CHECKS

Pivot, milling and RPM: checks B

Questions 17, 21, 22, 24, 25 · Circle one answer each. Read pages 27 to 29.

17. Why should students avoid copying a milling setup from another job?

- A. Every approved setup relates to the confirmed workpiece, drawing and demonstrated process
 - B. Every milling job uses different machine brands
 - C. Different setups always produce identical results
-

21. Before calculating drilling RPM, where should the cutting-speed and drill-diameter values come from?

- A. The teacher-provided data table for the nominated operation
 - B. A remembered value from a previous job
 - C. A similar drill size used by another student
-

22. Why should the units be checked before substituting values into the approved RPM formula?

- A. To make the calculation look neater
 - B. To ensure the calculation uses consistent, teacher-approved units
 - C. To avoid writing the formula
-

24. According to the theory, what does the calculated RPM represent?

- A. A theoretical spindle speed that still requires comparison with authorised settings
 - B. The exact drill-press setting to use immediately
 - C. The only acceptable speed for every drill press
-

25. A calculated RPM does not exactly match a drill-press speed. What should happen next?

- A. Select the closest setting without checking
 - B. Compare the result with available settings using the SOP and teacher demonstration
 - C. Repeat the calculation until the numbers match
-

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13.1 online Q1–10 | 13.2 online Q1–10 | 13.3 online Q1–10

INVESTIGATE / APPLY

Calculate, interpret, then confirm

Module 4 / Theoretical RPM is one part of the evidence chain.

Use the metric formula on page 29. Arithmetic practice only: a hypothetical cutting speed of 15 m/min and drill diameter of 12 mm are supplied below. These values do not specify a shovel material, drill or machine setting.

For the second task, name the data source and record v_c , D , their units, the approved formula, working, rounding direction, available setting and teacher confirmation. If the values or authorisation have not been provided, record what is pending instead of inventing a result.

Calculate the theoretical RPM for $v_c = 15 \text{ m/min}$ and $D = 12 \text{ mm}$. Show the formula, substitution and units. Round the arithmetic result to one decimal place, then explain why it is not automatically an authorised setting.

Present a fully labelled calculation using teacher-supplied values, then state how the authorised available setting is confirmed and recorded.

READ / OBSERVE / APPLY

Handle and pivot assembly as a functional quality gate

14.1 / Assembly and evaluation

Assembly is a functional quality gate: it brings separately made components together and tests their relationships against the issued drawing and criteria. Confirm the identity, orientation and intended position of each part before interpreting movement. A fault in the complete assembly may originate in an earlier feature or an incorrect relationship between otherwise tidy parts.

Use the teacher-approved check to observe whether components sit correctly and follow the controlled movement path without unexpected resistance, contact or misalignment. Where a thread or fastener relationship is confirmed by the drawing, inspect it by the authorised method. Force and appearance cannot replace evidence about fit or engagement.

Trial assembly keeps the work reversible while the checks occur. Repositioning or separating parts may be allowed by the demonstrated process. Rework is different: it changes a manufactured feature. A student must not treat an assembly difficulty as permission to remove material, alter a component or adopt a different fastener.

For example, an observation might locate unexpected contact during a teacher-approved movement check. Record the condition and reference, investigate possible reasons with suitable evidence, and identify the teacher-confirmed response. Keep the original observation in the record even after the issue is resolved. Photographs and annotations should explain what was checked and what remains uncertain. The student record supports review; the current criteria and teacher approval control readiness for finishing or handover.

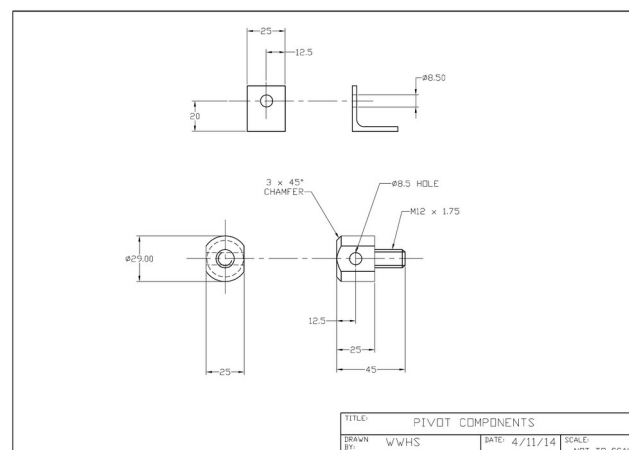


Figure 13. Source-sheet overview for locating related views. Read the complete, legible original on page 7; use its written information, not measurements from this overview.

Write an assembly quality-gate record covering confirmed criterion, movement evidence, issue diagnosis and teacher-confirmed action.

READ / OBSERVE / APPLY

Societal and environmental impact report with evidence

14.2 / Assembly and evaluation

The final impact report should evaluate what was actually made and recorded. Begin by separating project evidence from external information. A material list, production photograph or waste record describes this project. A credible external source may explain wider material or manufacturing effects. Each supports a different type of claim.

Societal impacts include confirmed user value, maintenance, repairability and possible service life. Environmental impacts include resource use, manufacturing waste, finishing and end-of-life options. Evidence for one category does not automatically prove another. A good-looking surface, for example, does not by itself establish durability or responsible material use.

Use a claim–evidence–limitation structure.

State a bounded claim, cite the relevant record or source, explain how it supports the claim, and acknowledge what it cannot establish. If durability is proposed as a way of reducing early replacement, evidence about quality and intended use is still needed. Do not invent a service life from a photograph.

A balanced conclusion makes trade-offs visible. A decision may improve serviceability while increasing material use or processing. Describe both effects and identify a realistic improvement linked to the records. Where evidence is missing, state the limit rather than filling it with an optimistic estimate. Correctly acknowledge external sources and distinguish a possible benefit from a verified outcome. The report should show why its conclusion is reasonable, not simply attach a label such as “sustainable”.



Figure 14. Material and process records support bounded claims about resources. Illustrative stock view; the report must use the actual project's records.

Write one supported impact paragraph containing claim, evidence/source, limitation and realistic improvement.

READ / OBSERVE / APPLY

PMI, project report and authentic Task 2 submission

14.3 / Assembly and evaluation

The final project report connects the evidence chain from brief to evaluation. Bring together the approved requirements and drawings, WMS, SOP evidence, cutting and costing list, production records, quality checks and impact report. A collection of disconnected activities does not yet explain how the product developed through decisions.

Use photographs as evidence, with captions that name the stage, explain what the image proves and connect it to a criterion or decision. Cost and component records must describe what was actually used and made. If actual use differs from the plan, record the change and its reason rather than editing history to look unchanged.

PMI means plus, minus and interesting. A plus identifies a supported success; a minus identifies a limitation; an interesting point identifies a significant finding or question. Explain each point's effect and, where evidence permits, its likely cause. An improvement should be achievable and connected to the finding rather than a vague promise to "do better".

For example, a recorded alignment check may help explain an observed assembly result; liking the appearance does not prove that relationship. Be honest about faults, assistance, missing evidence and pending approval. Acknowledge sources and teacher demonstrations. The current teacher-issued notice controls assessment timing, marks and submission. This workbook does not issue a new Task 2 notice or turn a completion checkbox into submission or practical approval.



Figure 15. An evidence chain connects source, observation, decision and evaluation. Illustrative planning scene; your report must describe your own as-built work.

Draft a concise final report conclusion that integrates quality, impact, cost/production evidence, limitation and next-step improvement.

MODULE 5 / SELECTED KNOWLEDGE CHECKS

Assembly and evaluation: checks A

Questions 3, 6, 9, 11, 15 · Circle one answer each. Read pages 33 to 35.

3. According to the theory, how should fit be assessed?

- A. By applying extra force until parts fit together.
 - B. By judging whether the assembly looks correct.
 - C. By using verified evidence rather than force or appearance.
-

6. Why is trial assembly completed before final finishing?

- A. To permanently join every component.
 - B. To avoid checking the drawing again.
 - C. To keep the work reversible while relationships between parts are checked.
-

9. When may a student declare the assembly safe or complete?

- A. As soon as all components are fitted together.
 - B. After the assembly looks correct.
 - C. Only after teacher approval using the confirmed acceptance criteria.
-

11. What should be the main basis for judging the completed shovel's impacts?

- A. Evidence from the completed project and credible referenced sources
 - B. General opinions about similar products
 - C. What usually happens in metalwork projects
-

15. Why should an impact report discuss trade-offs?

- A. Because every production decision has only positive effects
 - B. Because improvements in one area may increase impacts in another
 - C. Because trade-offs are only required for environmental impacts
-

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14.1 online Q1–10 | 14.2 online Q1–10 | 14.3 online Q1–10

MODULE 5 / SELECTED KNOWLEDGE CHECKS

Assembly and evaluation: checks B

Questions 16, 21, 23, 27, 28 · Circle one answer each. Read pages 33 to 35.

16. If information needed to support an impact claim is unavailable, what should the report do?

- A. Estimate a likely conclusion
- B. State the limitation instead of inventing a conclusion
- C. Ignore the missing information

21. What is the main purpose of the final PMI evaluation and project report?

- A. To consolidate the complete evidence chain for the project
- B. To introduce new production activities
- C. To replace the production records

23. A photograph in the folio is most useful when it includes a caption that:

- A. Explains the production stage, what the image proves and its connection to a decision or criterion
- B. Only names the tool shown
- C. States that the work looks good

27. Which statement best reflects authentic submission?

- A. Only successful work should be included
- B. Changes, assistance and unresolved approval should be acknowledged
- C. Every planned feature must be reported as completed

28. Where should you confirm unresolved assessment details such as submission directions or approval status?

- A. The current Task 2 handbook, teacher-issued documents and teacher directions
- B. Another student's folio
- C. Internet examples of similar projects

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14.1 online Q1–10 | 14.2 online Q1–10 | 14.3 online Q1–10

INVESTIGATE / APPLY

Build a defensible final judgement

Module 5 / Connect assembly evidence, impacts and evaluation.

A final report is more persuasive when a reader can trace the conclusion backwards to a criterion and evidence. Work from your actual records. Where a check is pending, identify it rather than writing the result you expect to obtain.

Choose one significant quality or impact conclusion. Complete an evidence chain: criterion or question → actual source/observation → interpretation → limitation → proposed improvement. Explain the connection at each step.

Audit the chain. Identify any claim that needs more evidence, any assistance that should be acknowledged and the teacher checkpoint still needed before final submission.

PROJECT FOLIO

Project brief and function

Folio 1 / Your source, decision and evidence.

Summarise the need, intended user, function and teacher-issued requirements.

Use the current brief to separate the user's need from your personal preferences. State the intended function, the confirmed user/context and the requirements that would demonstrate success. Do not add a capacity, load or operating claim that the teacher-issued sources do not establish.

Write a brief that links the need, user, function and confirmed requirements. Identify one requirement that still needs clarification.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Requirements and modifications

Folio 2 / Your source, decision and evidence.

Separate requirements from constraints and explain any teacher-approved modification.

A requirement states what the project must achieve; a constraint limits the available choices. A proposed modification becomes part of the project only when the teacher confirms it. Keep the original requirement and the reason for an approved change visible.

Record requirements and constraints, then explain the status of any proposed modification.

Requirement or constraint	Source and implication

Describe a proposed or approved modification, the evidence for it and its current approval status. If none is proposed, explain one requirement that should be preserved.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

WHS and risk controls

Folio 3 / Your source, decision and evidence.

Record hazards, risk level, stronger controls, PPE and teacher permission points.

Use the current SOP, WMS template, demonstration and teacher directions. The record should connect the actual stage to a hazard, confirmed control and permission point. A general PPE list does not explain how risk is controlled in this particular task.

Record two teacher-confirmed stages and their risk-control evidence.

Stage, hazard and risk level	Confirmed control / permission	Evidence and review

Explain why one confirmed control is appropriate and what changed condition would require the task to be reviewed.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Materials investigation

Folio 4 / Your source, decision and evidence.

Compare relevant metals or stock forms and justify the teacher-specified selection.

Compare materials or stock forms against the same requirements. Distinguish a material property from a shape or supply form. Use the teacher-specified material for the project; the comparison explains that choice and does not authorise a substitution.

Compare two relevant candidates or stock forms using verified evidence.

Criterion / evidence source	Candidate or form A	Candidate or form B

Justify the teacher-specified selection using function, manufacture and one relevant trade-off. Identify any missing material evidence.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

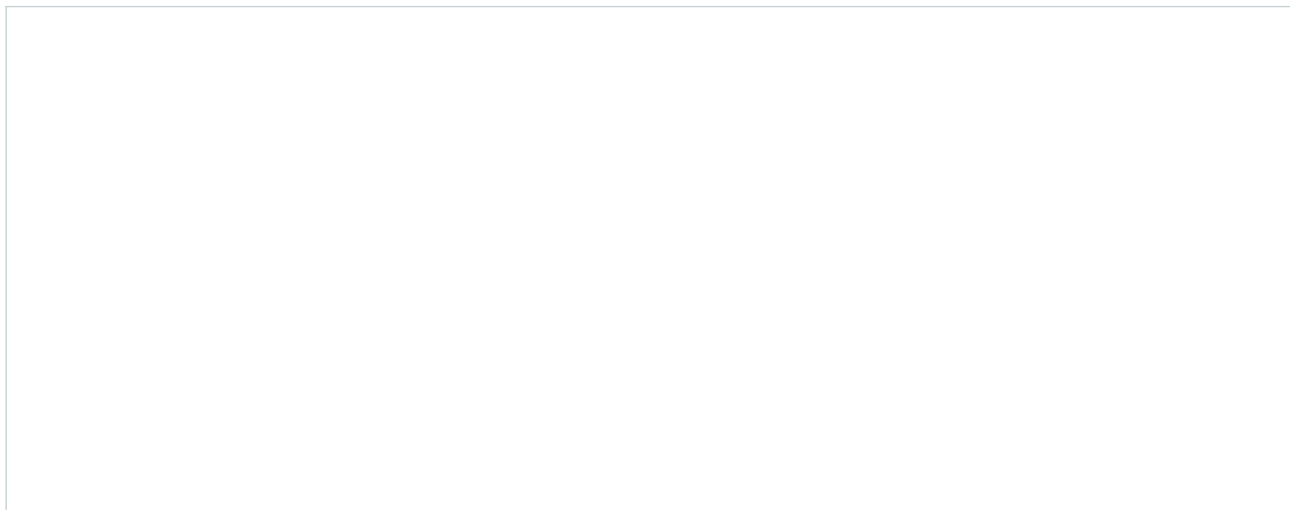
Research and ideas

Folio 5 / Your source, decision and evidence.

Record research, freehand sketches or modification ideas and explain the approved direction.

Record the source of an idea and what it helps you investigate. A commercial product photograph can suggest a question; it cannot replace the teacher-issued shovel drawings. Keep any design exploration clearly separate from an approved construction change.

Sketch or map one researched idea. Label the need or question it addresses, its source and the features that still need verification.



Explain why this idea is useful and state the teacher-confirmed direction or approval still required.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

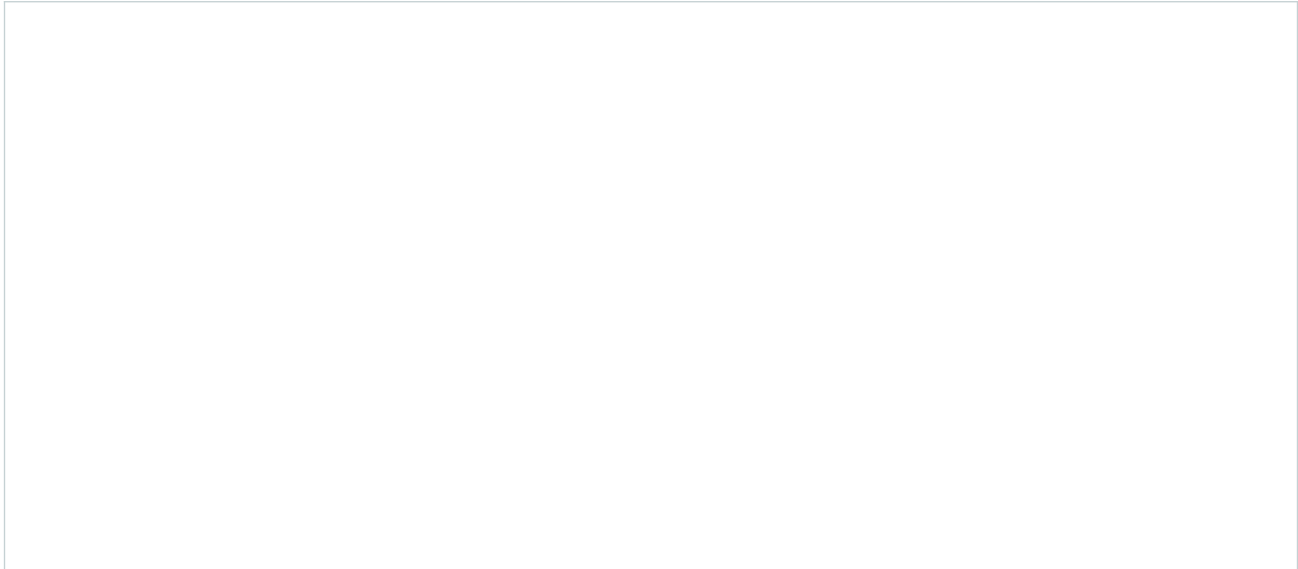
Working drawing evidence

Folio 6 / Trace the information from source to component.

Explain how the teacher-issued drawing communicates material, dimensions, shape, tolerance or assembly information.

Choose a component relationship that you can verify in the original sheets on pages 3–8. Use a freehand explanation sketch to connect the relevant views. Label the component, the views or sheet titles and the information each contributes. Copy only confirmed information; write “Teacher to confirm” beside a genuine uncertainty.

Component relationship sketch — identify the views and show how they connect.



Drawing explanation

Evidence caption — what does your sketch or authorised photograph demonstrate?

Source or teacher checkpoint — record sheet title, date and any clarification.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Cutting, costing and resources

Folio 7 / Your source, decision and evidence.

Summarise your spreadsheet list and explain one decision that reduced waste or improved efficiency.

Use the actual spreadsheet cutting and costing list. This paper record helps you explain it; it does not authorise stock dimensions or replace the current list. Record real quantities and costs from verified sources, and preserve the difference between planned and actual use.

Select three entries from your spreadsheet and show what each tells you about resources.

Component / spreadsheet row	Confirmed quantity / source	Cost or waste evidence

Explain one evidenced decision that reduced waste or improved efficiency. Identify the record that supports your conclusion.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Production plan and WMS

Folio 8 / Your source, decision and evidence.

Sequence key processes, evidence checkpoints, demonstrations and approvals.

Plan from the approved sequence and current WMS. Show dependencies: what later work relies on each result? Place a hold point where evidence or permission must be established before continuing. A table of tool names alone cannot explain the project plan.

Record two connected stages from the approved plan.

Stage / source reference	Dependency, control and hold point	Evidence / permission

Explain why the hold point belongs before the dependent stage, and how a changed condition would be recorded and reviewed.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Production evidence

Folio 9 / Your source, decision and evidence.

Write captions for your own photographs that explain the process, control and result shown.

Choose your own authorised photographs or observations. A caption should identify the stage, explain the control and show what the result establishes. Avoid photographs that expose other students or private information. An image is useful only when its limits are understood.

Plan or record two production evidence items. Attach authorised photographs separately if needed.

Stage / image or observation reference	Process, control and result shown

Explain one decision supported by this evidence and one thing the image or observation cannot prove by itself.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Quality and tolerance checks

Folio 10 / Your source, decision and evidence.

Record the approved criteria, measurement or fit evidence, result and any correction.

Select one meaningful quality check from your project. Separate the acceptance criterion from the actual evidence and the judgement. If the result is uncertain, record what must be checked next. A possible cause and a teacher-approved correction are separate parts of the record.

Create a traceable quality record.

Record element	Your project evidence
Confirmed criterion and source	
Measurement / observation and method	
Comparison and result	
Possible cause / authorised next action	

State what remains unresolved and the evidence or teacher review needed to close the check.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

Industry and impacts

Folio 11 / Your source, decision and evidence.

Compare one classroom and industrial process, then explain a societal or environmental impact.

Compare a classroom process with an industrial counterpart using the same basis, such as scale, consistency, resources or quality control. Then connect one societal or environmental claim to real evidence and a limitation. Wider research does not replace your own project records.

Compare one confirmed classroom process with a researched industrial counterpart.

Comparison basis	Classroom evidence	Industrial source

Write an impact paragraph with a bounded claim, evidence/source, explanation, limitation and realistic improvement.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.

PROJECT FOLIO

PMI and final evaluation

Folio 12 / Your source, decision and evidence.

Use plus, minus and interesting observations to judge function, quality and process, then identify one realistic improvement.

PMI records plus, minus and interesting observations. Connect each to evidence and an effect, then propose an improvement that follows from the findings. Acknowledge help, changes and missing evidence. Use the current teacher notice for submission requirements.

Complete an evidence-based PMI.

PMI	Observation, effect and supporting evidence
Plus	
Minus	
Interesting	

Conclude your report: judge function, quality and process, name a limitation and justify one realistic improvement.

Evidence caption — what does this record or authorised photograph demonstrate?

Source or teacher checkpoint — identify the source used and any confirmation.

[] Evidence checked and ready. Optional authorised photo may be attached to your evidence record.